

Mid Term Exam CIT 315: Artificial Intelligence
 Total Marks: 15 Duration: 40 minutes

1. How can an agent be rational? Explain.	3
2. Discuss different types of agents.	3
3. Explain the criteria to evaluate search strategies.	4
4. Compare uninformed search strategies.	5

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Patuakhali Science and Technology University
 Mid Term Exam
 Course Code: CCE 310 Course Title: Software Development Project-I
 Total Marks: 15 Time: 30 minutes

- Write a simple Windows Forms MessageBox statement.
- Write the connection string for database connection in C# and explain each parameter.
- State the insert, select, delete and update SQL command.
- Explain the function of Reader class in C#.

2
3
6
4

Mid Term Exam of Microprocessors and Assembly Language(CIT-311), Session: 2015-2016
 Time: 40 Min. Marks: 15

- Which physical address is represented by: EFFF:1234H
 - What will be the CS:IP of physical address BCDEFh where CS=FFF0?
- Describe the content of segment register at protected mode memory addressing. Distinguish between real and protected mode memory addressing.
- Define program invisible register. Give the flag status of flag register after performing the following operation.

ABCD	
	×A
- Mention address lines of 8086. How does queue of 8086 speed up the processing?

Course Code: CCE 313 Course Title: Computer Networks Time: 45 minutes

- What is the network address in a block of addresses? How can we find the network address if one of the addresses in a block is given? 3
- An ISP is granted a block of addresses starting with 120.60.4.0/22. The ISP wants to distribute these blocks to 100 organizations with each organization receiving just eight addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations. 5
- In a connection, the value of *cwnd* is 2500 and the value of *rwnd* is 4500. The host has sent 2000 bytes which has not been acknowledged. How many more bytes can be sent? 2
- A window holds bytes 2001 to 6000. The next byte to be sent is 3001. Draw a figure to show the situation of the window after the following two events: 4
 - An ACK segment with the acknowledgement number 3500 and window size advertisement 4000 is received.
 - A segment carrying 1500 bytes is sent.
- What is a socket address? 1

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions. Answer must be **brief, relevant and neat**.

- 1 [A.] Explain about the learning outcomes for studying the Computer Organization and Architecture course. 3
- [B.] Illustrate the basic elements of a digital computer. 2
- [C.] What are the roles of different register of control unit and the ALU? 3
- [D.] How are data read from a magnetic disk? 3
- [E.] List and briefly define three techniques for performing I/O. 3

- 2 [A.] Four benchmark programs are executed on three Laptops with the following results: 3

	Laptop A	Laptop B	Laptop C
Program 1	10	10	20
Program 2	1000	100	20
Program 3	500	1000	50
Program 4	100	800	100

The table shows the execution time in seconds, with 200,000,000 instructions executed in each of the four programs. Calculate the MIPS values for each computer for each program. Then calculate the arithmetic and harmonic means assuming equal weights for the four programs, and rank the computers based on arithmetic mean and harmonic mean.

- [B.] Consider the below computer specification 5
- Pentium Intel® Core™ i7 4.7 GHz
 - 2GB 1333 MHz DDR3 SoDIMM expandable Up to 4GB memory
 - 32 KB L1 Cache, 256 KB L2 cache and 4MB L3 Cache
 - 1TB 7200 RPM SATA8 Hard drive
 - (6) High speed USB 2.0 (2 side/4 rear), (2) Side audio ports: headphone and microphone, (2) PS/2 ports, Serial2
 - 19" Inches LCD Monitor, 0.25 mm AG, 1280×1024 at 80 MHz
 - 48x DVD RW Optical
 - 256 MB PCI express graphics card
 - 56K data/fax Modem
 - 64 bit PCI sound card
 - Intel® 82578DM, 10M/100M/1000M Gigabit Ethernet
 - What does it all mean?

- [C.] Discuss about cache/main memory structure with block diagram 3
- [D.] Your computer Hard disk drive with 8 surfaces, 512 tracks per surface, and 64 sectors per track. Sector size is 1 KB. The average seek time is 8 ms, the track-to-track access time is 1.5 ms, and the drive rotates at 3600 rpm. Successive tracks in a cylinder can be read without head movement. 3

- i) What is the disk capacity?
- ii) What is the average access time? Assume this file is stored in successive sectors and tracks of successive cylinders, starting at sector 0, track 0, of cylinder i
- iii) Estimate the time required to transfer a 5-MB file.
- iv) What is the burst transfer rate?

- 3 [A.] What are the basic elements of Cache Design? 2

[B.] What is the general relationship among access time, memory cost, and capacity?

[C.] A set-associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses.

[B.] Explain Direct-Mapping Cache Organization with diagram

[B.] Distinguish between DRAM and SRAM in terms of characteristics such as application, speed, size, and cost.

4 [A.] Explain hardware implementation of unsigned binary multiplication for multiplicand=13 and multiplier=11.

[B.] Analyze the statement "CPU performs all arithmetic operation using addition".

[C.] i) Define instruction format. Why does the variable length instruction format used in CPU operation?

ii) Mention the concept of von Neumann architecture.

[D.] Justify the statement- "More dedicated interconnection produce faster computer". Differentiate between system bus and local bus.

5 [A.] i) What is the difference between the two's complement representation of a number and two's complement of a number?

ii) If an instruction contains four address, what might be the purpose of each address?

[B.] i) What are the typical elements of machine instruction?

ii) What types of locations can hold source and destination operands?

[C.] A hypothetical machine has 6 instructions

0010=Load AC from memory	1011=Sub memory from AC
0011=Store AC to memory	0110=Store AC to I/O
0101=Add to AC from memory	1001=Load AC from I/O

In this case, the 12-bit address identifies a particular I/O device. Show the program execution (using 16 bit instruction format where prefix 4 bits for Opcode) for the following program.

- Load AC from memory location 999.
- Add contents of memory location 555.
- Store AC to device 6.

Assume that the next value retrieved from memory location 999 is 2 and location 555 contains a value of 3.

[D.] List and explain the instruction set design issues.

[A.] Explain two's complement division procedure for dividend $Q = -7$ and divisor $M = -3$.

[B.] Analyze the relationship between instruction and micro-operation.

ii) Explain performance and performance penalty of instruction pipelining.

[C.] Explain RISC architecture. Differentiate between RISC and CISC.

[D.] i) Differentiate between arithmetic shift and logical shift.

ii) A given microprocessor has words of one byte. What is the smallest and largest integer that can be represented in the following representation?

- Unsigned
- Two's complement

2. a) Determine the normal equations for polynomial of the nth degree 6
b) Find the values of a, b and c so that $Y = a + bx + cx^2$ is best fit to the data 4

x	y
0	1
1	0
2	3
3	10
4	21

- c) Fit a function of the form $y = ax^b$ to the following data 4

x	y	x	y
2	43	20	8
4	25	40	5
7	18	60	3
10	13	80	2

3. a) Derive the LU decomposition method for solving a system of n linear equations 7

b) Solve the following system 5

$$5x - 2y + z = 4$$

$$7x + y - 5z = 8$$

$$3x + 7y + 4z = 0$$

By the unit lower triangular and upper triangular (LU) method.

- c) Write down the difference between Gauss elimination method and Gauss-Jordan method. 2

3. a) Explain the term 'interpolation' 3

- b) Deduce Gauss's interpolation formula 6

- c) Using Gauss's forward formula, find the value of $f(32)$ given that $f(25)=0.2707$, $f(30)=0.3027$, $f(35)=0.3386$, $f(40)=0.3794$ 5

4. a) Discuss the method of Bisection to find an approximate root of an equation $f(x) = 0$ 6
b) Find a real root of the equation $x^2 - x - 2 = 0$ by using Bisection Method and False position method. 4+4=8

5. a) Describe the Taylor Series Method for the solution of ordinary differential equation. 6

Given $\frac{dy}{dx} = 1 + xy$ when $y(0) = 2$ find $y(0.1)$ by Taylor Series Method and Runge

- b) Kutta 2nd order method. 4+4=8

6. a) Derive general integration formula to compute $\int_a^b f(x)dx$ and Simpson's 1/3 rule for the numerical integration. 3+4=7

- b) Compute the integral $\int_0^1 \frac{dx}{\sqrt{1+x^2}}$ upto 4 decimal place by using Trapezoidal rule and Simpson's 1/3. 3+4=7

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Faculty of Computer Science and Engineering
Mid Term Examination of B.Sc.Engg (CSE) January-June 2011
Course Title : Microprocessors and Microcomputers
Course Code : CSE-311
Level : III, Semester : I (Session : 2008-09).

Time : 1 Hour

Marks : 20

N.B. Answer the following questions.

1. (a) Why microprocessor so called?
- (b) What do you mean by logical and physical address? How can you determine the physical address from logical. ✓
- (c) Describe about the flag register? Give the status of a flag register of 8086 after performing the following operation. ✓

BCDEh
+ 9876h

2. (a) Briefly describe about the memory addressing mode of processor? Is the real addressing mode used in modern processor? 4 ✓
- (b) "A particular memory segment always start with a fixed location" Justify this statement. 2 ✓
- (c) How can we obtain 16-bit machine language instruction from assembly language instruction? 4 ✓

Ajay Uddin
d. Hemant

Faculty of Computer Science and Engineering
Mid Term Examination of B.Sc. Engg. (CSE) January-June 2010
Course Title : Microprocessors and Microcomputers
Course Code : CSE-311
Level : III, Semester : I (Session : 2007-08).

Time : 50 Min

Marks : 20

N.B. Answer the following questions.

1. (a) What do you mean by n bit processor? ✓

(b) Suppose CS register contains 368A and IP register contains 368A. Determine the address depending on both CS and IP. ✓

(c) Why does the processor use flag register? Give the status of a flag register after performing the following operation.

$$\begin{array}{r} 1111 \\ +1101 \\ \hline \end{array}$$

(d) Why the handshaking procedure is used in computer system? #

2. (a) What do you mean by the volatile and non-volatile memory? O

(b) We know that 8086 is a 16 bit microprocessor. But its memory mapping is 20 bits. How it is possible?

(c) Describe about the protected and real addressing mode.

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

1. a) Explain the basic functions of a computer. 4
 b) Distinguish between computer architecture and organization. Explain why do you study computer architecture and computer organization. 5
 c) What are the basis of von Neumann architecture? Enlist the registers of IAS computer with their functionality. 5
 Explain Moore's law. 2
2. a) List and explain the key characteristics of a computer family. 4
 b) Explain the possible states that define an instruction execution. Briefly define two approaches to dealing with multiple interrupts. 5
 c) Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder contains the immediate operand or an operand address. 5
 - i) What is the maximum directly addressable memory capacity (in bytes).
 - ii) Discuss the impact on the system speed if the microprocessor bus has a 32-bit local address bus and 16-bit local data bus.
 - iii) How many bits are needed for the program counter?
3. a) Why is interconnections necessary? Explain control signals of computer system. 4
 b) What are the considering elements of bus design? Describe synchronous bus operation. 5
 c) How can you characterize the memory of your computer? Distinguish among direct, associative and set associative mapping of cache. 5

$$A \quad M \quad M-1 \quad Q(-17)$$
4. a) Calculate the multiplication operation using Booth's algorithm for $Q = -17$ (multiplicand) and $M = 16$ (multiplier). 4
 b) Explain set associative mapping. For hexadecimal main memory addresses FFFFFFFF, ABD9EF, show the following information in hexadecimal format. (Where cache size is 64 KBytes) 5
 - i) Tag, Line and Word values for a direct mapped cache.
 - ii) Tag, Set and Word values for two-way set associative cache.
5. a) Why is the cache memory necessary in computer system? Describe the replacement algorithms of cache memory. 5
5. a) i) What is the general relationship among access time, memory cost, and capacity? 4
 ii) Assume numbers are represented in 6-bit two's complemented representation. Show calculation of the following.
 - a) $-21+7$
 - b) $-6-13$
- b) i) What is the relationship between instructions and micro-operation? Suppose your ALU can add its two input registers. List the micro-operations your control unit must perform to cause an addition. 5
 ii) Distinguish between RISC and CISC processor.
- c) What is RAID? Explain error detection and error correction mechanisms in semiconductor memory. 5
6. a) What is the difference between arithmetic shift and logical shift. 4
 b) i) What general roles are performed by processor registers? 5
 ii) Explain peripheral devices with its classification. 5
 c) i) Define I/O modules with its major functionality. 5
 ii) What are the typical elements of a machine instruction?
 iii) List and explain the instruction set design issue.
 iv) Shortly describe variable length addressing? "Each operation needs operand"-justify this statement.

3, 5, 6,
2189

5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

Patuakhali Science and Technology University
 B.Sc. Engg. (CSE) Level-3, Semester-1 Final Examination-2014 (January - June)
 Course Code: CIT-313 Course Title : Computer Architecture
 Credit Hour : 3.0 Full Marks:70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

- 1) Establish a relationship between computer architecture and organization. Explain basic function of a computer. 7
- 2) What do you mean by stored computer? Enlist the registers of IAS computer with its functionality. Explain about Moore's law. 7
- 3) Explain interrupt with its classification. Why does the interrupt used in computer system? How does the processor handle multiple interrupts? 7
- 4) Explain two's complement division procedure for dividend $Q = -13$ and divisor $M = -11$. Assume numbers are represented in 8-bit two's complement representation. Show calculation of the following. 7
 - i. $-10-1$
 - ii. $11-17$

- 5) How can you characterize a memory? Explain direct mapping for cache memory. Why replacement algorithm is necessary in cache? 7
- 6) Explain the design issues of instruction set. For hexadecimal main memory addresses 111111, ABCDEF, show the following information in hexadecimal format, (Where cache size is 64 KBytes) 7
 - a. Tag, Line and Word values for a direct mapped cache.
 - b. Tag, Set and Word values for two-way set associative cache.

- 7) Distinguish among direct, associative and set associative mapping of cache. Explain multiplication algorithm for multiplicand $Q = 17$ and multiplier $M = 19$. 7
- 8) A hypothetical machine has 6 instructions 7

0000=Load AC from memory	0010=Sub memory from AC
0001=Store AC to memory	0111=Store AC to I/O
0101=Add to AC from memory	1111=Load AC from I/O

In this case, the 12-bit address identifies a particular I/O device. Show the program execution (using 16 bit instruction format where prefix 4 bits for opcode) for the following program.

- i. Load AC from memory location 999.
- ii. Add contents of memory location 555
- iii. Store AC to device 6.

Assume that the next value retrieved from memory location 999 is 2 and location 555 contains a value of 3

IF
OF
FI
FI

- 9) What general roles are performed by CPU registers? Explain different addressing techniques in computer programming. 7
- 10) Describe the role of condition codes with its advantages and disadvantages. Differentiate between parallel access and independent access of RAID. How does the RISC design philosophy impacted on technology trends? 7
- 11) Why does bus timing essential? How does the program control the hardware? Distinguish between hardware controlled device and micro-programmed controlled device. 7
- 12) Describe pipelining. Suppose your ALU can add its two input registers, and it can logically complement the bits of either input register, but it cannot subtract. Numbers are to be stored in two's complement representation. List the micro-operations that control unit must perform to cause a subtraction. 7

3. a) Derive the Gauss elimination method for a system of n linear equations and hence write the difference between Gauss method and Gauss-Jordan method. 7

b) Solve the following system 7

$$5x - 2y + z = 4$$

$$7x + y - 5z = 8$$

$$3x + 7y + 4z = 10$$

by (i) Gauss elimination method and (ii) Gauss-Jordan method.

4. a) Derive the Bisection Method and False position method. 8

b) Find a real root of the equation $(x^3 + x^2 - 1 = 0)$ by using False position Method. 6

5. a) Derive the Trapezoidal rule and Simpson's 1/3 rule. 7

b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Trapezoidal rule and Simpson's 1/3. 7

6. a) Describe the Runge Kutta 2nd order method for the solution of first order ordinary differential equation. 7

b) Solve $\frac{dy}{dx} = x + y$, $y(0) = 1$ and find $y(0.2)$ by Runge Kutta 4th order method. 7

[Figures in the right margin indicate full marks. Split answering of any question is highly discouraged. Write the full question number e.g. 1(B)(i) before the answer paragraph]

Answer any 5 of the following questions

- 1 A Use Gauss-Elimination technique to solve the following system 7

$$\begin{aligned} 0.2x_1 - 0.7x_2 - 0.2x_3 &= 7.85 \\ 0.10x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.30x_1 - 0.2x_2 + x_3 &= 71.4 \end{aligned}$$
- 1 B i. How can we remove the error of Trapezoidal rule by applying Simpson's rule? 7
 ii. Sketch and state graphical depiction of singular and ill-conditioned systems.
- 2 A Show a case where Bisection is preferable to False Position. 7
 2 B i. State Weddle's Rule. 7
 ii. Solve the following system using Cramer's rule

$$\begin{aligned} 0.3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.7x_1 + 0.7x_2 - 0.3x_3 &= -19.3 \\ 0.30x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$
- 3 A Apply Factorization process to find out the root of the following system 7

$$\begin{aligned} x_1 + x_2 - x_3 &= 2 \\ 2x_1 + 3x_2 + 5x_3 &= -3 \\ 3x_1 + 2x_2 - 3x_3 &= 6 \end{aligned}$$
- 3 B i. Write down the algorithm of Bolzano's Method. 7
 ii. Use Simpson's 3/8 rule to integrate
 $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $a=0$ to $b=0.8$
- 4 A Show the effect of reduced step size on Euler's method. 7
 B i. Show two cases of iteration process. 7
 ii. What is Lagrange's interpolation formula? In which case we use it?
- 5 A "Secant method can be classified as a bracketing method" --- is it true or false? 7
 Explain your answer clearly.
- 5 B i. Develop equation for linear regression. 7
 ii. Show that $x_r = x_u - f(x_u)(x_l - x_u) / (f(x_l) - f(x_u))$ in case of false position method.
- 6 A Use N-R method to estimate the root of $f(x) = e^{-x} - x$, employing an initial guess of $x_0 = 0$ (True value = 0.56714329) until $\epsilon_t(\%) < 10^{-8}$. 7
- 6 B Use Gauss-Jordan technique to solve the following system 7

$$\begin{aligned} 3x_1 - x_2 - 0.2x_3 &= 7.85 \\ 0.10x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.30x_1 - 0.2x_2 + x_3 &= 71.4 \end{aligned}$$

Patuakhali Science and Technology University
5th Semester (L-3, S-1), Final Exam. of B. Sc. Engg. (CSE), January-June, 2015

Course Code: CCE-313 Course Title: Computer Networks

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Give the answers of any 5 questions from the given questions:

1. (a). Discuss about the protocols in computer networking. 6
~~(b). Discuss about bus, ring, star and mesh topology with cable link connection of the given topology.~~ 8
2. (a). What are the major duties of data link layer? How data link layer completes node-to-node delivery? 8
(b). Write down the short note on connection control, flow control and error control in transport layer. 6
3. (a). Distinguish a unicast address from multicast address. 4
~~(b). Discuss the functions of 802.3 MAC frame.~~ 6
~~(c). Why bridged and switched Ethernet are used in networking systems?~~ 4
4. (a). Show the encoding of 1000 base-X and 1000 base-T Ethernet. 6
~~(b). What are the features of RS and MII of fast Ethernet?~~ 4
~~(c). Why repeater is used in networking system? Compare the performance of repeater and hub.~~ 4
5. (a). What are the loop problems for bridge connections? Show the example of loop problems 6
(b). Show the steps of spanning tree in a bridged LAN 6
(c). How does a VLAN reduce network traffic? 2
6. (a). Discuss about store and forward packet switching in network layer. 4
~~(b). How each node is labeled in shortest path routing algorithm?~~ 3
~~(c). Why subnet masks are used in computer networking?~~ 3
~~(d). In a computer network, Network address is given as 196.64 .10.0
Subnet Mask is given as 255.255.255.248~~ 4

Find the answer of following questions

- i. What are the valid subnets?
- ii. What are the valid first hosts?

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Which topology in computer networks is better according to your perception? Show your justifications. 4
- (b). Write down the responsibilities of data link layer and transport layer in internet model. 10
2. (a). Discuss about the MAC frame in Traditional Ethernet. 7
- (b). Write down the short note on RS, GMII, PHY, and MDI in Gigabit Ethernet. 7
3. (a). What is ARP? Discuss about the fields in ARP packet. 8
- (b). Find out the checksum in the given IP packet 6

4	5	0	28
1	0	0	
4	17	1	
10.12.14.5			
12.6.7.9			

4. (a). Implement the connection-oriented service in the network layer. 6
- (b). What is datagram subnet? Implement the datagram subnet. 8
5. (a). Show how distance vector routing works in a subnet. 6
- (b). What is routing algorithm? Discuss about the shortest path routing algorithm. 8
6. (a). Show the timing of events in circuit switching and packet switching. 4
- (b). In a computer network,
 Network address is given as 192.168.10.0
 Subnet Mask is given as 255.255.255.248
 Find the answer of following questions 10

- i. How many subnets?
- ii. How many hosts?
- iii. What are the valid subnets?
- iv. What are the valid hosts?
- v. What are the broadcast addresses for each subnet?

Faculty of Computer Science and Engineering
Patuakhali Science and Technology University

Final Examination of B.Sc. Engineering in CSE Level: 3 Semester: I Session: 2015-16
Course Title: Numerical Methods
January-June 2018
Credit: 03
Time: 03 Hr
Marks: 70
Course Code: CCE 311
Answer any 05 out of 06 Questions (Split answers are highly discouraged)

Put appropriate figure/example if necessary

- 1 A Determine the real roots of $f(x) = -0.4x^2 + 2.2x + 4.7$ 2+4
Using the quadratic formula
Using three iterations of the bisection method to determine the highest root. Employ initial guesses of lower $x_1 = 5$ and $x_0 = 10$. Compute the estimated error e_a and true error e_t after each iteration. 2+3
- 1 B Explain accuracy and precision with example. Write down different types of ^{error} found in numerical analysis with the formula to calculate them. 3
- 1 C Derive the formula of false position method with proper figure. 3+3
- 2 X How do you use Taylor series in the derivation of Newton-Raphson method? Explain the pitfalls of Netwan-Raphson method with example. 5
- 2 B Use simple fixed-point iteration to locate the root of $f(x) = -0.9x^2 + 1.7x + 2.5$
Use an initial guess of $x_0 = 5$ and iterate until $e_a \leq 0.01\%$ 3
- 2 X State the differences between secant and false-position method. 6
- 3 A Use Mullar's method with guesses of x_0, x_1 and $x_2 = 4.5, 3.5$ and 5.5 respectively, to determine a root of the equation
 $f(x) = x^3 - 13x - 12$
Note that the roots of this equation are -3, -1, and 4 5
- 3 B Apply Cholesky decomposition to the symmetric matrix 1.5+1.5
 $[A] = \begin{bmatrix} 6 & 15 & 35 \\ 15 & 55 & 225 \\ 35 & 225 & 979 \end{bmatrix}$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$
- 3 Q Write short notes on
i. LU decomposition
ii. Lower and Upper triangular matrix

$$\begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix}$$

Fit a second-order polynomial to the data of the following table. Also find out standard error $S_{y/x}$

x_i	y_i
1	7.7
2	13.6
3	27.2

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- 4 B Use Simpson's 3/8 rule to integrate
 $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $a=0$ to $b=0.8$
Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments
Find out E_t for both cases. true val = 1.640533
- 5 A Derive equation for linear regression and show criteria for a "Best" fit. 7
- 5 B Given $dy/dx = \frac{1}{2}(x+y)$, $y(0)=2$, $y(0.5)=2.636$, $y(1.0)=3.595$, $y(1.5)=4.968$. Find $y(2)$ by Milne's method. 7
- 6 A Use the point-slope method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from $X=0$ to $X=4.0$ with a step size 0.5. The initial condition at $X=0$ is $Y=1$. 7
- 6 B Given $dy/dx = 1+xy$ and $y(0)=1$. Calculate $y(0.1)$, $y(0.2)$ using Picard's method. 7

$$\frac{y^2}{2} + 12 \cdot \frac{y^3}{3} - 20 \cdot \frac{y^4}{4} + 8.5y$$

Patuakhali Science and Technology University

Semester (Level-3, Semester-I) Final Examination of B.Sc. Engg. (CSE)-2018 (January-June)
Course Code: CIT-311 Course Title : Microprocessors and Microcomputers
Credit Hour : 3.00 Session: 2015-16 Full Marks:70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
Answer any 5 of the following questions.

- 1 a) Explain 16 bit instruction format of 8086. What will be the CS: IP of physical address FBCDEh in 8086? 2+2
- b) For an 80486 descriptor that contains a base address of 010000Fh, and a limit of 001FFh. What starting and ending locations are addressed by this descriptor considering G=0 and G=1? 3
- c) Describe following instructions with respect to 8086. 4
 - i) MOV 43H[SI], BX
 - ii) MOV CS:[AX], DI
- d) "CPU actually works on binary digits"- Justify this statement. 3
- 2 a) Why is memory decoding needed in computer system? Describe 3-state buffer basic input interface. 2+2
- b) Define n-bit processor. 8086 is a 16 bit microprocessor, but its memory mapping is 20 bits. How it is possible? 3
- c) What is descriptor? Describe the access rights byte of descriptor. 2+2
- d) How does the pointer register work with their corresponding segment registers? 3
- 3 a) Why is DMA used in computer system? Explain trigger of DMA. 2+2
- b) Describe addressing modes of DSP56300. 3
- c) Give the responsibility coprocessor in computer system? Mention 6 processors with its corresponding coprocessors. 2+2
- d) Explain the priority between a DMA channel and the core. 3
- 4 a) Distinguish between microcontroller and microprocessor. Give the application of microcontroller. 2+2
- b) Give the features of 8031 microcontroller. Enlist the brands of microcontroller. 2+1
- c) Write short notes with key features of 2+2
 - i) Pentium iv
 - ii) Core i5
- d) Explain floating gate transistor with respect to EPROM. 3
- 5 a) Describe the responsibility of memory management unit in computer system. Determine the number of page table entries that are needed for following combinations of virtual address size (n) and page size (P). 2+2

n	P=2 ^p	#PTE
16	4K	
64	16K	
- b) What is virtual address? What do you mean by pages in virtual memory system? 1+2
- c) Define handshaking. Distinguish between 8085 and 8086. 2+2
- d) Explain "how does onionskin operating system provide resource protection". 3
- 6 a) Why is PPI used in computer system? Explain the basic operation of TSR program. 2+2
- b) Compare among mode 0, mode 1 and mode 2 of 82C55PPI. 3
- c) How does the page fault handle in virtual memory system? 4
- d) Describe the role of operating system in virtual memory. 3

FB 208
EF 298

EF 210, 2A 9E
3

0170087807

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

1. Establish a relationship between computer architecture and organization. Explain basic function of a computer. 7
2. What do you mean by stored computer? Enlist the registers of IAS computer with its functionality. Explain about Moore's law. 7
3. Explain interrupt with its classification. Why does the interrupt used in computer system? How does the processor handle multiple interrupts? 7
4. Explain two's complement division procedure for dividend $Q = -13$ and divisor $M = -11$. Assume numbers are represented in 8-bit two's complement representation. Show calculation of the following. 7
 - i. $-10-1$
 - ii. $11-17$
5. How can you characterize a memory? Explain direct mapping for cache memory. Why replacement algorithm is necessary in cache? 7
6. Explain the design issues of instruction set. For hexadecimal main memory addresses 111111, ABCDEF, show the following information in hexadecimal format, (Where cache size is 64 KBytes) 7
 - a. Tag, Line and Word values for a direct mapped cache.
 - b. Tag, Set and Word values for two-way set associative cache.
7. Distinguish among direct, associative and set associative mapping of cache. Explain multiplication algorithm for multiplicand $Q=17$ and multiplier $M=19$. 7
8. A hypothetical machine has 6 instructions 7

0000=Load AC from memory	0010=Sub memory from AC
0001=Store AC to memory	0111=Store AC to I/O
0101=Add to AC from memory	1111=Load AC from I/O

In this case, the 12-bit address identifies a particular I/O device. Show the program execution (using 16 bit instruction format where prefix 4 bits for opcode) for the following program.

- i. Load AC from memory location 999.
- ii. Add contents of memory location 555
- iii. Store AC to device 6.

Assume that the next value retrieved from memory location 999 is 2 and location 555 contains a value of 3.

9. What general roles are performed by CPU registers? Explain different addressing techniques in computer programming. 7
10. Describe the role of condition codes with its advantages and disadvantages. Differentiate between parallel access and independent access of RAID. How does the RISC design philosophy impacted on technology trends? 7
11. Why does bus timing essential? How does the program control the hardware? Distinguish between hardware controlled device and micro-programmed controlled device. 7
12. Describe pipelining. Suppose your ALU can add its two input registers, and it can logically complement the bits of either input register, but it cannot subtract. Numbers are to be stored in two's complement representation. List the micro-operations that control unit must perform to cause a subtraction. 7

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

1. Explain the basic functions of a computer. 4
2. Distinguish between computer architecture and organization. Explain why do you study computer architecture and computer organization. 5
3. What are the basis of von Neumann architecture? Enlist the registers of IAS computer with their functionality. 5
4. Explain Moore's law. 4
5. List and explain the key characteristics of a computer family. 4
6. Explain the possible states that define an instruction execution. Briefly define two approaches to dealing with multiple interrupts. 5
7. Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder contains the immediate operand or an operand address.
 - i) What is the maximum directly addressable memory capacity (in bytes).
 - ii) Discuss the impact on the system speed if the microprocessor bus has a 32-bit local address bus and 16-bit local data bus.
 - iii) How many bits are needed for the program counter?
8. Why is interconnections necessary? Explain control signals of computer system. 4
9. What are the considering elements of bus design? Describe synchronous bus operation. 5
10. How can you characterize the memory of your computer? Distinguish among direct, associative and set associative mapping of cache. 5
11. Calculate the multiplication operation using Booth's algorithm for $Q = -17$ (multiplicand) and $M = 16$ (multiplier). 4
12. Explain set associative mapping. For hexadecimal main memory addresses FFFFFFFF, ABD9EF, show the following information in hexadecimal format. (Where cache size is 64 KBytes) 5
 - i) Tag, Line and Word values for a direct mapped cache.
 - ii) Tag, Set and Word values for two-way set associative cache.
13. Why is the cache memory necessary in computer system? Describe the replacement algorithms of cache memory. 5
14. a) i) What is the general relationship among access time, memory cost, and capacity? 4
 Assume numbers are represented in 6-bit two's complement representation. Show calculation of the following.
 - a) $-21+7$
 - b) $-6-13$
15. b) i) What is the relationship between instructions and micro-operation? Suppose your ALU can add its two input registers. List the micro-operations your control unit must perform to cause an addition. 5
16. ii) Distinguish between RISC and CISC processor. 5
17. c) What is RAID? Explain error detection and error correction mechanisms in semiconductor memory. 5
18. a) i) What is the difference between arithmetic shift and logical shift. 4
19. ii) What general roles are performed by processor registers? 5
20. b) i) Explain peripheral devices with its classification. 5
21. ii) Define I/O modules with its major functionality. 5
22. c) What are the typical elements of a machine instruction? 5
23. d) List and explain the instruction set design issue. 5
24. e) (iii) Shortly describe variable length addressing? "Each operation needs operand"-justify this statement.



[A] **Topology Orientation and Building a Small Network**

10

Task 1: Create a Peer-to-Peer Network.

Task 2: Identify the Cables used in a Network.

Task 3: Cable the Peer-to-peer Network.

Step 1: Connect two PC.

Step 2: Apply a Layer 3 address to the PC

Step 3: Verify connectivity

Task 4: Use the ping command to verify that PC1 can reach PC2 and PC2 can reach PC1.

Step 5: Share a document between PCs.

Task 5: Group/file name

2.

Question 1

According to the OSI model, which layer addresses the issues of communication media and cable connectors etc?

- a. Network
- b. Data Link
- c. Transport
- d. Physical

Question 2

Which cable uses RJ45 connectors?

- a. Thin Coaxial Cable
- b. Thick Coaxial Cable
- c. Unshielded Twisted Pair Cable
- d. Shielded Twisted Cable

Question 3

Which cabling media does not suffer from Electro-Magnetic Interference (EMI)?

- a. Thin Coaxial Cable
- b. Thick Coaxial Cable
- c. Unshielded Twisted Pair Cable
- d. Fibre Optic Cable

Question 4

Which layer is broken down into the MAC and LLC sub-layers?

- a. Network
- b. Data Link
- c. Session
- d. Physical

Question 5

What is the purpose of the shielding in a Shielded Twisted Pair (STP) cable?

- a. To reduce attenuation
- b. To reduce cross talk
- c. To reduce Elector-Magnetic Interference

Question 6

Which of the following is a Class A network?

- a. 172.0.0.0
- b. 126.0.0.0
- c. 192.0.0.0

d. 225.0.0.0

10*0.5=5

Question 7

According to the OSI model, at which layer are network addresses assigned?

- a. Session
- b. Data Link
- c. Presentation
- d. Network

Question 8

At which layer does a Hub operate?

- a. Network
- b. Physical
- c. Session
- d. Data link

Question 9

According to the OSI model, which layer performs the function of transforming bits of data into electrical signals and sending those signals across the media?

- a. Network
- b. Data Link
- c. Transport
- d. Physical

Question 10

How is a unique MAC address assigned to a Network Card?

- a. It is built into the card when the card is manufactured
- b. The network administrator must assign the address
- c. A unique address is automatically assigned when the card drivers are installed
- d. A unique address is automatically assigned whenever you boot up the computer

N.B: If wrong answer then will cut 0.25marks for every question.

Mid: I
Course Code: CCE 313

Time: 45 Min Course Title: Computer Networks

- 4
1. [A.] Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.
- Express the propagation delay, d_{prop} , in terms of m and s .
 - Determine the transmission time of the packet, d_{trans} , in terms of L and R .
 - Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
 - Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?
 - Suppose d_{prop} is greater than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
 - Suppose d_{prop} is less than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
 - Suppose $s = 2.5 \times 10^8$, $L = 120$ bits, and $R = 56$ kbps. Find the distance m so that d_{prop} equals d_{trans} .
- [B.] List six access technologies. Classify each one as home access, enterprise access, or wide-area wireless access. 2
- [C.] What is an application-layer message? A transport-layer segment? A network layer datagram? A link-layer frame? 3
- [D.] An e-commerce site hatbazar.com that wants to keep a purchase record for each of its customers. Explain how this can be done with cookies. 3
- [E.] Let's walk through the steps of transferring a Web page from server to client for the case of non-persistent connections. The webpage page consists of a base HTML file and 12 JPEG images, and that all 13 of these objects reside on the same server. The URL for the base HTML file is <http://www.pstu.ac.bd/someDepartment/home.index> 3

- [A.] Two different machines A and B, with two different instruction sets, both of which have a clock rate of 200 MHz. The following measurements are recorded on the two machines running a given set of benchmark programs: 4

Instruction Type	Instruction Count (millions)	Cycles per Instruction
Machine A		
Arithmetic and logic	8	1
Load and store	4	3
Branch	2	4
Others	4	3
Machine B		
Arithmetic and logic	10	1
Load and store	8	2
Branch	2	4
Others	4	3

- ii. Determine the effective CPI, MIPS rate, and execution time for each machine.
 iii. Comment on the results.
- [B.] Why do you study Computer Architecture and Organization? 2
 [C.] Explain the expanded structure of IAS computer. 4
 [D.] Why interrupts are necessary in computer system? Shortly explain different types of interrupts. 3
 [E.] Explain the responsibility of different registers for executing an instruction. 2

1. a) Evaluate $\int_0^1 \frac{dx}{\sqrt{1+x^2}}$ By Simpson's 1/3 rule. 4

- b) Compute $y(0.5)$ and $y(1)$ using ~~R-K~~ ^{Euler's} 2nd order method for $\frac{dy}{dx} = \frac{1}{x+y}$ with i.c $y(0) = 1$ 3

2. a) Find the best values of a_0, a_1 and a_2 so that the parabola $y = a_0 + a_1x + a_2x^2$ fits the data: 4

x	y
1.0	1.1
1.5	1.2
2.0	1.5
2.5	2.6
3.0	2.8
3.5	3.3
4.0	4.1

- b) Fit a function of the form $y = ax^b$ to the following data: 4

x	y
2	43
4	25
7	18
10	13
20	8
40	5
60	3
80	2

character that was contained at that point in the HTTP header). Answer the following questions, indicating where in the HTTP GET message below you find the answer.

```
GET /CCE313/index.html HTTP/1.1<cr><lf>Host: pstu.ac.bd<cr><lf>User-Agent: Mozilla/5.0 (Windows;11;
Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax) cr><lf>Accept:ex t/xml, application/xml
application/xhtml+xml, text/html;q=0.9, text/plain;q=0.8,image/png,*/*;q=0.5
<cr><lf>Accept-Language: en-us,en;q=0.5<cr><lf>AcceptEncoding: zip,deflate<cr><lf>Accept-Charset: ISO-
8859-1,utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr><lf>Connection:keep-alive<cr><lf><cr><lf>
```

Course Title: Microprocessors and Assembly Language
Credit Hours:3.00, Course Code : CIT-311, Session:2013-2014

Time: 45 Min.

Marks: 15

N.B. Answer the following questions. (Split answers are highly discouraged)

- 1 Is RAM a peripheral? Explain your answer. 2
- 2 Traditional computer system has 3-category(data, address, control) buses. If a designer tries to design a computer by using more or less than 3-category buses, explain the problems for this case. 3
- 3 Shortly mention the evolutionary scenario of computing device. Also describe the working procedure of EPROM and EEPROM. 4
- 4 Why is the protected mode necessary in computer? How does the descriptor table identify itself as global or local? 3
- 5 Explain the role of segment register. Suppose CS=FFF0 and IP=FFFF, calculate the memory address with respect to 8086. 3

Patuakhali Science and Technology University
Mid Term Examination-2016

Course Code: CIT-315 Course Title: Artificial Intelligence

Full Time: 40 minutes

Full Marks: 15

1. Describe components of a neural network and discuss the similarities of neural network with human brain. 6
2. What does "learning" mean referring to neural nets? 4
3. Differentiate between supervised and unsupervised learning. 2
4. Explain forward propagation and self-organization. 3

Faculty of Computer Science and Engineering

Final Examination of B.Sc.Engg. (CSE) January-June 2012

5th Semester (Level-3, Semester-I), Session: 2009-2010

Course Title: Microprocessors and Microcomputers /

Credit Hours: 3.00, Course Code : CSE-311

Time: 3 Hours

Marks: 70

N.B. Answer any seven questions from the followings. (Split answers are highly discouraged)

1. (a) Define n -bit processor and handshaking? *Page: 2.4 #* 3
- (b) Why does the processor use flag register? Give the status of flag register of 8086 after performing the following operation. *Page: 2.12* 3

A00C	90 972
-0FFD	90 031
2. (a) "Firmware is software"-Justify this statement. Mention the role of the firmware in computer system. 4
- (b) What is the difference between 8086 and 80286? *Page: 2.11* 3
- (c) Suppose an application program requires more memory than available primary memory. How this problem could be solved? 3
- (d) Describe protected mode memory addressing. Distinguish between real and protected mode memory addressing. 4
3. (a) Why DMA is used in computer? Enlist the data transfer types of a DMA controller. * 3
- (b) Distinguish between RISC and CISC processor. *unq. shift* 3
- (c) Why memory management is necessary in multitasking system? Explain how to manage the memory in multitasking system? ③ 4
4. (a) Distinguish between RAM and Cache. *2.10* 3
- (b) What do you mean by IN and OUTS? Why stepper motor is discrete? *Memory interface* 3
- (c) Shortly describe 82C55 PPI. How it can be programmed? ④ 4
5. (a) How memory segments are overlapped in real addressing mode? 3
- (b) What is general purpose register? Why AX is called accumulator? *Page: 2.13* 3
- (c) Assembly language programming is comparatively easy than machine language-how can we obtain a 16-bit machine language instruction of corresponding assembly language instruction? *Page: 2.16* 4
6. (a) How the individual segment register can be used in CPU operation. 3
- (b) How DMA controller handles the unmatched source and destination? * 3
- (c) Describe 80287. Why coprocessor is used in computer system? ① 4
7. (a) "All coprocessors are peripherals, but all peripherals are not coprocessors"-Explain this statement. ① 3
- (b) Distinguish among PROM, EPROM, and EEPROM. ① 3
- (c) "EPROM is a nonvolatile one time programmable memory"- Justify this statement and describe your judgment. 4
- Or 0
8. (a) Describe about virtual memory addressing. ②
9. (a) Why memory management unit is necessary in computer system? ③ 3
- (b) Write short note on any two from the followings. 7
 - i) Pentium 4
 - ii) Dual core
 - iii) Core i3

Apu
Ria
Rifat
Shorrel
Tareen

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
Answer any 5 of the following questions.

1. (a) Give the comparative description of the hardware controlled device and the microprogram controlled device. 3
(b) Differentiate between corei3 and corei5. 3
(c) "Segment register can locate any memory cell as the base address of the respective segment among the whole memory" describe this statement with examples. 4
(d) Why coprocessor is used in the computer system? How does the coprocessor interface with the main processor? 4
2. (a) Describe about two layer onionskin operating systems. 3
(b) Distinguish among PROM, EPROM, and EEPROM. 3
(c) "CPU actually works on binary digits"- Explain this statement. 4
(d) Continuous motor is familiar to us, but stepper is a digital motor. Why it is digital? How it can be operated through 82C55 PPI? 4
3. (a) Differentiate between dual core and core 2 duo. 3
(b) Briefly describe about memory overlay mechanism. 3
(c) For an 80486 descriptor that contains a base address of 01000000h, and a limit of 001FFh. What starting and ending locations are addressed by this descriptor considering G=0 and G=1? 4
(d) Describe about SSD. 4
4. (a) Distinguish between 80286 and 8086. 3
(b) How does the 82C55 can be programmed? 3
(c) Why memory management is necessary in multitasking system? How can you manage the memory in multitasking system? 4
(d) Why the protected mode memory addressing is used in the computer system? Distinguish between real and protected addressing mode. 4
5. (a) "All coprocessors are peripherals, but all peripherals are not coprocessors"-Explain this statement. 3
(b) Distinguish between global descriptor and local descriptor. 3
(c) Why address decoding is necessary? Shortly describe one technique for this purpose. 4
(d) Why DMA is used in computer? Explain DMA triggering. * ✓ 4
6. (a) Suppose an application program requires more memory than the available primary memory. How does this problem could be solved? 3
(b) Briefly describe the operation modes of 82C55. 3
(c) Describe following instructions with respect to 8086. 6
 i) MOV CL, [AX]
 ii) MOV 43H[SI], AX
 iii) MOV AX, [437AH]
 iv) MOV CS:[AX], DI
(d) Mention address lines of 80286. What will be the CS: IP of physical address ABCDEh in 8086? 2

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
Answer any 5 of the following questions.

1. a) "CPU actually works on binary digits"- Explain this statement. 3
 b) How does the DMA controller handle the unmatched source and destination? 3
 c) Why the processor use the protected mode memory addressing and how does the processor use this mode? 4
 d) Distinguish between real and protected addressing mode. 4
 e) 8086 has numerous instructions. Some instructions can be grouped depending on their functional similarity. Briefly describe them. 4
2. a) The pointer registers [SP, BP, IP] are in CPU, How does these pointer registers work with their corresponding segment registers? 3
 b) Why does the handshaking procedure used in computer system? 4
 c) "Segment register can locate any memory cell as the base address of the respective segment among the whole memory" describe this statement with examples. 4
 d) Distinguish between 8086 and 80286. What will be the CS:IP of physical address ABCDEh in 8086? 4
3. a) Define n -bit processor. Give the status of flag register of 8086 after performing the following operation. 3
 F0C9
 - 0AFD
 b) 8086 is a 16 bit microprocessor, but its memory mapping is 20 bits. How is it possible? 3
 c) For an 80486 descriptor that contains a base address of 01000000h, and a limit of 001FFh. What starting and ending locations are addressed by this descriptor considering G=0 and G=1? 4
 d) Why does the coprocessor used in computer system? Enlist the data transfer types of a DMA controller. 4
4. a) Why AX is called accumulator? What is the role of AX register in CPU operation? 3
 b) Define coprocessor. Briefly explain about DMA triggering. 3
 c) Shortly describe 82C55 PPI. How it can be programmed? 4
 d) Describe following assembly language instructions with respect to 8086. 4
 i) MOV CL, [AX]
 ii) MOV AX, [437AH]
 iii) MOV CS:[AX], DI
5. a) Describe about two layer onionskin operating systems. 3
 b) Briefly explain about IN and OUTS. 3
 c) Why memory management unit is necessary in computer system? 4
 d) EPROM is a nonvolatile one time programmable memory". Justify this statement and describe your judgment. 4
6. a) Give the comparative discussion of intel dual core over intel pentium processor. 3
 b) Describe about core i5 or core i7. 3
 c) What do you mean by multitasking? How can we achieve this in our computer system? 4
 d) Why virtual memory is used in computer? Explain the virtual memory system of computer. 4

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I) Final Examination of B.Sc. Engg. (CSE) (January-June 2018)

Course Code: CIT-313 Course Title : Computer Organization and Architecture

Credit Hour : 3.00 Session: 2015-16 Full Marks:70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions. Answer must be brief, relevant and neat.

1. a) A hypothetical machine has 6 instructions

1010=Load AC from memory	1011=Sub memory from AC
0011=Store AC to memory	0110=Store AC to I/O
0101=Add to AC from memory	1101= Load AC from I/O

In this case, the 12-bit address identifies a particular I/O device. Show the program execution (using 16 bit instruction format where prefix 4 bits for opcode) for the following program.

- Load AC from device 99.
- Add contents of memory location 555
- Store AC to memory location 666.

Assume that the next value retrieved from ^{device} memory location 99 is 2 and location 555 contains a value of 3.

- Distinguish between computer architecture and organization. 2+4
 - Mention the computer generation with its corresponding technology. What general roles are performed by processor registers?
- Why do we need to study computer organization and architecture? 2

- Explain the distinction between the written sequence and time sequence of an instruction. Illustrate and list the micro-operations to perform addition between two operands from memory. 2+4
 - Differentiate between CISC and RISC architectures. What are their typical characteristics? Give some examples of processors of each category. 2×3
 - Why error correction is needed in memory system? 2

- Mention the key features of RAID 0-RAID 6. Pipeline processor is inefficient for branch instruction. Explain the mechanisms to deal the branches in pipelining. 3+3
 - Assume a pipeline with 4 stages: fetch instruction (FI), decode instruction and calculate address (DA), fetch operand (FO), and execute (EX). Draw a timing diagram for instruction pipeline operation for a sequence of 9 instructions, in which the fifth instruction is a branch that is taken and in which there is no data dependencies. 4+2
 - Mention data types for multimedia and give 2 multimedia instruction of Intel Pentium product.
 - Mention the advantages and disadvantages of variable length addressing. 2

- Describe the elements of a machine instruction. Assume numbers are represented in 6-bit two's complemented representation. Show calculation of the following. 3+3
 - A+B (Hex)
 - 33-17 (Decimal)

- Find out the quotient and remainder by performing twos complement division operation where divisor = -3 and dividend = -7. 4+2
- Illustrate the arithmetic shift operation.
- Enlist different addressing mode. 2

- Why Input/output module is used in computer system? Describe the functions of I/O module. 2+4
 - Shortly explain peripheral devices with its classification. Distinguish between program driven and interrupt driven I/O. 2+4
 - Distinguish between SRAM and DRAM. 2

- Compare direct, associative and set-associative memory. Explain replacement algorithms of cache memory. 3+3
 - Mention the control signals among system module. Explain multiple interrupt with examples. 3+3
 - Give the characteristics of computer family. 2

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

- 1 a) Explain 16 bit instruction format of 8086. What will be the CS: IP of physical address FBCDEh in 8086? 2+2
 b) For an 80486 descriptor that contains a base address of 0100000Fh, and a limit of 001FFh. What starting and ending locations are addressed by this descriptor considering G=0 and G=1? 3
 c) Describe following instructions with respect to 8086. 4
 i) MOV 43H[SI], BX
 ii) MOV CS:[AX], DI → DI to memory. E
 d) "CPU actually works on binary digits"- Justify this statement. 3
- 2 a) Why is memory decoding needed in computer system? Describe 3-state buffer basic input interface. 2+2
 b) Define *n*-bit processor. 8086 is a 16 bit microprocessor, but its memory mapping is 20 bits. How it is possible? 3
 c) What is descriptor? Describe the access rights byte of descriptor. 2+2
 d) How does the pointer register work with their corresponding segment registers? 3
- 3 a) Why is DMA used in computer system? Explain trigger of DMA. 2+2
 b) Describe addressing modes of DSP56300. 3
 c) Give the responsibility coprocessor in computer system. Mention 6 processors with its corresponding coprocessors. 2+2
 d) Explain the priority between a DMA channel and the core. 3
- 4 a) Distinguish between microcontroller and microprocessor. Give the application of microcontroller. 2+2
 b) Give the features of 8031 microcontroller. Enlist the brands of microcontroller. 2+1
 c) Write short notes with key features of 2+2
 i) Pentium iv
 ii) Core i5
 d) Explain floating gate transistor with respect to EPROM. 3
- 5 a) Describe the responsibility of memory management unit in computer system. Determine the number of page table entries that are needed for following combinations of virtual address size (n) and page size (P). 2+2

n	P=2 ^P	#PTE
16	4K	
64	16K	
- b) What is virtual address? What do you mean by pages in virtual memory system? 1+2
 c) Define handshaking. Distinguish between 8085 and 8086. 2+2
 d) Explain "how does onionskin operating system provide resource protection". 3
- 6 a) Why is PPI used in computer system? Explain the basic operation of TSR program. 2+2
 b) Compare among mode 0, mode 1 and mode 2 of 82C55PPI. 3
 c) How does the page fault handle in virtual memory system? 4
 d) Describe the role of operating system in virtual memory. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) 5th Semester Level-3, Semester-I, Final Examination January-June/2018

Course code: CIT-315

Course Title: Artificial Intelligence

Credit hour: 3.00

Full marks: 70

Duration: 3 hours

Session: 2015-16

[Figures in the right margin indicate full marks.]

Answer any 5 of the following questions. Split Answering is not recommended.

1. a. Define Artificial Intelligence within 4 different paradigms with examples. 3
~~b. What is rationality? Differentiate between human agent and artificial agent in terms of rationality. 1+2~~
~~c. "Do not measure the performance of an agent based on its behaviour. Measure the performance of an agent based on what is wanted in the environment." – justify these statements. 4~~
~~d. Explain the steps that a problem solving agent uses to solve a problem with an example. 4~~
2. a. What are the components of a solution to a well-defined problem? Explain with an example. 3
~~b. Define a game. Differentiate between games and search problems. 1+2~~
~~c. Demonstrate minimax algorithm and alpha-beta pruning with examples. 3+3~~
~~d. Discuss an example of the application of game theory to a real life scenario. 2~~
3. ~~a. Analyse BFS technique in terms of performance measure. Provide reasons to every measure. 3~~
~~b. "Exponential time or space complexity is very expensive" – justify with an example in terms of time and memory requirements. 3~~
~~c. Compare iterative deepening search (IDS) and breadth first search (BFS) in terms of performance measure. Which one is preferred when the search space is large and the depth of the solution is unknown? Explain with an example. 2+2~~
~~d. Consider the following scenario for 8-puzzle game. Figure 3.d.1 is the initial state and figure 3.d.2 is the goal state. Which of the uninformed search technique would be best suited? Justify your answer in terms of performance measure 4~~
4. ~~a. How does greedy best first search work? Show using figure 1. What are its properties? 2+2~~
~~b. Demonstrate how greedy best-first search is incomplete using figure 1. 4~~
~~c. Demonstrate the working principle of A* search to find a path from Oradea to Bucharest (figure 1) 4~~
~~d. Define: Admissible Heuristic, Optimally Efficient. 1+1~~
5. ~~a. Compare supervised and unsupervised learning. Explain the structure of artificial neuron. 2+2~~
~~b. Why fuzzy logic is used in neural network? Explain the neural network with its classification. 2+3~~
~~c. Explain fuzzy inference system. Distinguish between digital computer and neural networks. 3+2~~
6. ~~a. Explain member function with its features. Distinguish between crisp set and fuzzy set. 3+2~~
~~b. Define defuzzification. Explain the importance of neural network with respect to artificial intelligence. 1+3~~
~~c. Consider the following linguistic variables and membership functions (figure 6.c.1) with respect to fuzzy logic. Calculate the speed of driving at 65% Cloud Cover and 37 F° 5~~

8		6
5	4	7
2	3	1

Figure 3.d.1

	1	2
3	4	5
6	7	8

Figure 3.d.2

Rules	Linguistic-variables
i) If it's sunny and cool drive fast	i) Temp: {freezing, cool, warm, hot} (Fig.: i)
ii) If it's overcast and freezing drive slow	ii) Cloud Cover: {overcast, partly cloudy, sunny} (Fig.: ii)
	iii) Speed: {slow, fast} (Fig.: iii)

$$f(n) = g(n) + h(n)$$

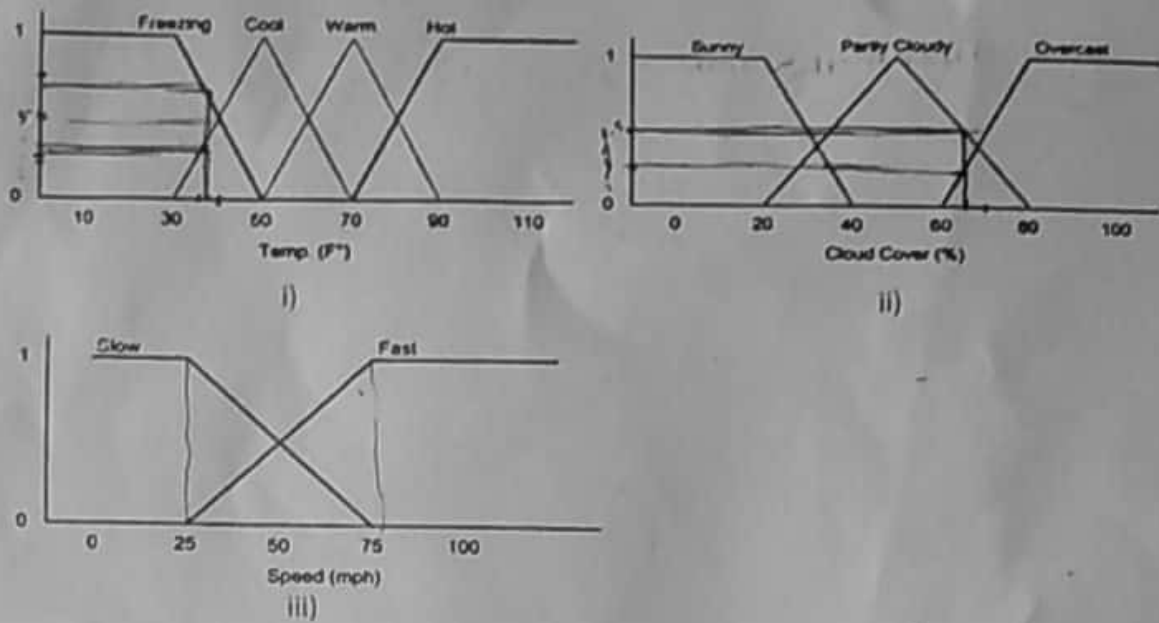


Figure 6.c.1

Romania with step costs in km



Figure 1

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

- 1 a) The following is a dump of a TCP header in hexadecimal format. 7
05320017 07000001 00000000 500207FF 00000000
 a. What is the source port number?
 b. What is the destination port number?
 c. What the sequence number?
 d. What is the acknowledgment number?
 e. What is the length of the header?
 f. What is the type of the segment?
 g. What is the window size?
- Source Port: 17
 Destination Port: 7
 Sequence: 05320017
 Acknowledgment: 07000001
 Header Length: 20
 Window Size: 00000000
 Offset: 500207FF
 Reserved: 00000000
- b) An ISP is granted a block of addresses starting with 120.60.4.0/22. The ISP wants to distribute these blocks to 100 organizations with each organization receiving just eight addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations. 5
- c) Why do we need a DNS system when we can directly use an IP address? 2
- 2 a) What is NAT? How can NAT help in address depletion? 3
- b) Briefly define subnetting and supernetting. How do the subnet mask and supernet mask differ from a default mask in classful addressing? 5
- c) A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. Find out the first address, last address, and the number of addresses in the block. Also show the network configuration for the block. 6
- 3 a) What is CIDR? Find the netid and the hostid of the following IP addresses: 5
 i. 217.34.38.0
 ii. 211.23.1.6
 iii. 114.56.7.83
 iv. 207.24.12
- b) In TCP, if the value of HLEN is 1000, how many bytes of option are included in the segment? 2
- c) A TCP connection is using a window size of 12,000 bytes, and the previous acknowledgement number was 22,001. It receives a segment with acknowledgement number 24,001 and window size advertisement of 12,000. Draw a diagram to show the situation of the window before and after. 5
- 4 a) In the standard Ethernet, if the maximum propagation time is 25.6 μ s, what is the minimum size of the frame? 2
- b) Illustrate the *Ethernet Frame*. Also show the minimum and maximum length. 4
- c) Write short notes on:
 i. Wireless LAN
 ii. Point-to-point WAN
 iii. Switched WAN 6
- d) How are options negotiated in TELNET? 4
- 5 a) Show a request that retrieves the document /usr/users/doc/doc1. Use at least two general headers, two request headers, and one entity header. Also show the response for a successful request. 5
- b) Write down the differences among repeaters, bridges, and routers. 4
- c) What is the purpose of the inverse domain? 2
- d) Define three types of web documents? Active, static, dynamic 3
- 6 a) When both sender and receiver are connected to the mail server via a LAN or a WAN what types of agents are needed and how many? Describe the situation. 7
- b) Differentiate between circuit switching and packet switching. 3

Final Examination of B.Sc. Engineering in CSE Level: 3 Semester: I Session: 2015-16

Course Code
CCE 311

Course Title
Numerical Methods

January-June
2018

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

Put appropriate figure/example if necessary

1. A Determine the real roots of $f(x) = -0.4x^2 + 2.2x + 4.7$ 2+4
 i. Using the quadratic formula
 ii. Using three iterations of the bisection method to determine the highest root. Employ initial guesses of lower $x_1 = 5$ and $x_4 = 10$. Compute the estimated error e_a and true error e_t after each iteration.

1. Explain accuracy and precision with example. Write down different types of ^{error} in numerical analysis with the formula to calculate them. 2+3

1. Derive the formula of false position method with proper figure. 2+3

2. How do you use Taylor series in the derivation of Newton-Raphson method? Explain the pitfalls of Netwan-Raphson method with example. 3+3

2. Use simple fixed-point iteration to locate the root of $f(x) = -0.9x^2 + 1.7x + 2.5$ 5
 Use an initial guess of $x_0 = 5$ and iterate until $e_a \leq 0.01\%$

2. State the differences between secant and false-position method. 3

3. Use Mullar's method with guesses of x_0, x_1 and $x_2 = 4.5, 3.5$ and 5.5 respectively, to determine a root of the equation $f(x) = x^3 - 13x - 12$ 6
 Note that the roots of this equation are -3, -1, and 4

3. Apply Cholesky decomposition to the symmetric matrix 5
 $[A] = \begin{bmatrix} 6 & 15 & 35 \\ 15 & 55 & 225 \\ 35 & 225 & 979 \end{bmatrix}$

3. Write short notes on 1.5+1.5
 i. LU decomposition
 ii. Lower and Upper triangular matrix

4. Fit a second-order polynomial to the data of the following table. Also find out standard error S_{yx} 7

x_i	y_i
1	7.7
2	13.6
3	27.2

4. B Use Simpson's 3/8 rule to integrate 7
 $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $a=0$ to $b=0.8$
 Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments
 Find out E_t for both cases. 7

5. A Derive equation for linear regression and show criteria for a "Best" fit. 7

5. B Given $dy/dx = \frac{1}{2}(x+y)$, $y(0)=2$, $y(0.5)=2.636$, $y(1.0)=3.595$, $y(1.5)=4.968$. Find $y(2)$ by Milne's method. 7

6. A Use the point-slope method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from $X=0$ to $X=4.0$ with a step size 0.5. The initial condition at $X=0$ is $Y=1$. 7

6. B Given $dy/dx = 1+xy$ and $y(0)=1$. Calculate $y(0.1)$, $y(0.2)$ using Picard's method. 7